Diagnostic Gateway ECU - HLD v1

Title and Document Control

SYNTHETIC DATA FOR ACADEMIC PROJECT

Diagnostic Gateway ECU - High Level Design

Document Version: v1

Gateway Components

DiagnosticGatewaySWC routes controlled diagnostic traffic between CAN0 and the diagnostic Ethernet uplink. DiagnosticManagerSWC handles diagnostic sessions inside the Vehicle Control ECU.

Interfaces

DiagnosticRequestInterface carries diagnostic requests. DiagnosticResponseInterface carries diagnostic responses. DiagnosticGatewaySWC requires DiagnosticRequestInterface and provides DiagnosticResponseInterface.

Dependencies

DiagnosticGatewaySWC depends on DiagnosticManagerSWC for application-level diagnostic processing and on CAN0 for vehicle-side transport.

Security and Review

Diagnostic execution must occur in an approved test environment. This synthetic architecture does not represent a production vehicle.